

How does excitation synchronize two PING networks?

exp085 · 19 August 2026 · Draft

Abstract

Two frequency-detuned PING networks are connected by reciprocal excitatory projections. At the selected demonstration point, direct E-to-E coupling arrests relative-phase drift. E-to-I coupling produces a narrow inhibitory-doublet response but does not lock the networks by itself.

Prior art

Lowet et al. used two 80-excitatory, 20-inhibitory PING networks connected through E-to-E and E-to-I projections. They found that locking depends mainly on frequency detuning and coupling strength. Their topology provides the starting point for this experiment: Lowet et al. (2015).

Phase-response analysis provides a way to connect individual coupling events to network-level locking. It also warns that a population rhythm can respond differently from any one neuron: Lowet et al. (2017) and Xu and Rieke (2021).

Methods

Methods 1–4 are complete. Method 5 is not yet run.

1. **Define the networks.** Build two matched PING networks. Each contains 80 excitatory and 20 inhibitory neurons connected through local E-to-I and I-to-E synapses. Drive Network A at 300 Hz and Network B at 260 Hz; these values place both uncoupled rhythms in the gamma band. Use a 1 ms local E-to-I AMPA decay so each inhibitory neuron fires once, rather than twice, after an excitatory volley. Connect the networks reciprocally through E-to-E and E-to-I projections. Each target receives exactly 8 randomly selected excitatory afferents from the other network. Give the pathways separate weights K_{EE} and K_{EI} and a shared delay d .

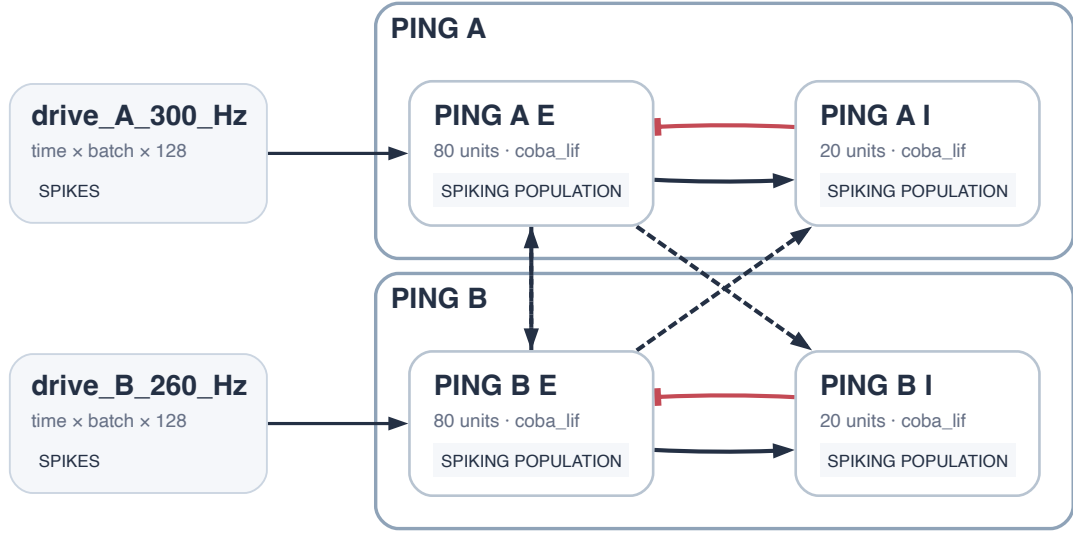


Figure 1: SNNLANG circuit export. Each network contains a local E-to-I-to-E PING loop. Reciprocal cross-network excitation targets both populations with exact fan-in 8, separate K_{EE} and K_{EI} controls, and shared delay d .

Generates Result 1.

2. **Confirm uncoupled phase drift.** Set $K_{EE} = K_{EI} = 0$ and run both networks for 2 s with independent input streams. Discard the first 300 ms. Detect excitatory population volleys, interpolate phase between successive volleys, and calculate the wrapped phase difference. Require both rhythms to remain in the gamma band with an inter-volley-interval coefficient of variation below 0.2. Generates Result 2.
3. **Measure each pathway's phase response.** Use Network A and one baseline cycle. A preliminary scan at intervals of 0.1 cycle located an abrupt inhibitory response near phase 0.1. The reported run therefore samples phases 0.02–0.30 at intervals of 0.02 and retains intervals of 0.1 over the rest of the cycle.
 Keep $K_{EE} = 0.08$, $K_{EI} = 0.08$, and $d = 2$ ms fixed. Only the probe target and arrival phase change. For desired phase φ , emit the source volley at $t_E + \varphi T - d$ so it reaches the target at $t_E + \varphi T$. Every condition uses the same network, initial state, input, and seed. Compare the next excitatory volley with the no-probe baseline. For I-targeted probes, also count inhibitory volleys and measure any second-volley latency. In a representative doublet, record local and probe excitatory conductance and every I-cell voltage. Positive phase shifts mean an advance; negative shifts mean a delay. Generates Result 3.
4. **Test the coupling pathways.** Run both networks uncoupled for 500 ms. Save their complete runtime state, then branch into four conditions: no coupling, E-to-E only, E-to-I only, and both pathways. Every branch starts from the same voltages, refractory states, live synapses, delayed events, and future input. Unwrap relative phase after coupling onset. Classify the final 500 ms as locked when absolute drift is below 0.25 cycles/s and phase concentration exceeds 0.95. Generates Result 4.

5. **Trace the mechanism.** Align activity on incoming excitatory volleys. Measure target excitatory activity, target inhibitory activity, inhibition received by the target excitatory population, and the phase correction in the following cycle. Generates Result 5.

Results

Result 1: Network definition

The authored graph contains two matched 80-excitatory, 20-inhibitory PING circuits. It keeps inhibition local and uses reciprocal long-range E-to-E and E-to-I projections. The two pathways have separate parameters, $K_{EE} = 0.08$ and $K_{EI} = 0.08$, with delay $d = 2$ ms. SNNLANG represents the sparse projections through exact-fan-in initialization.

Result 2: Uncoupled phase drift

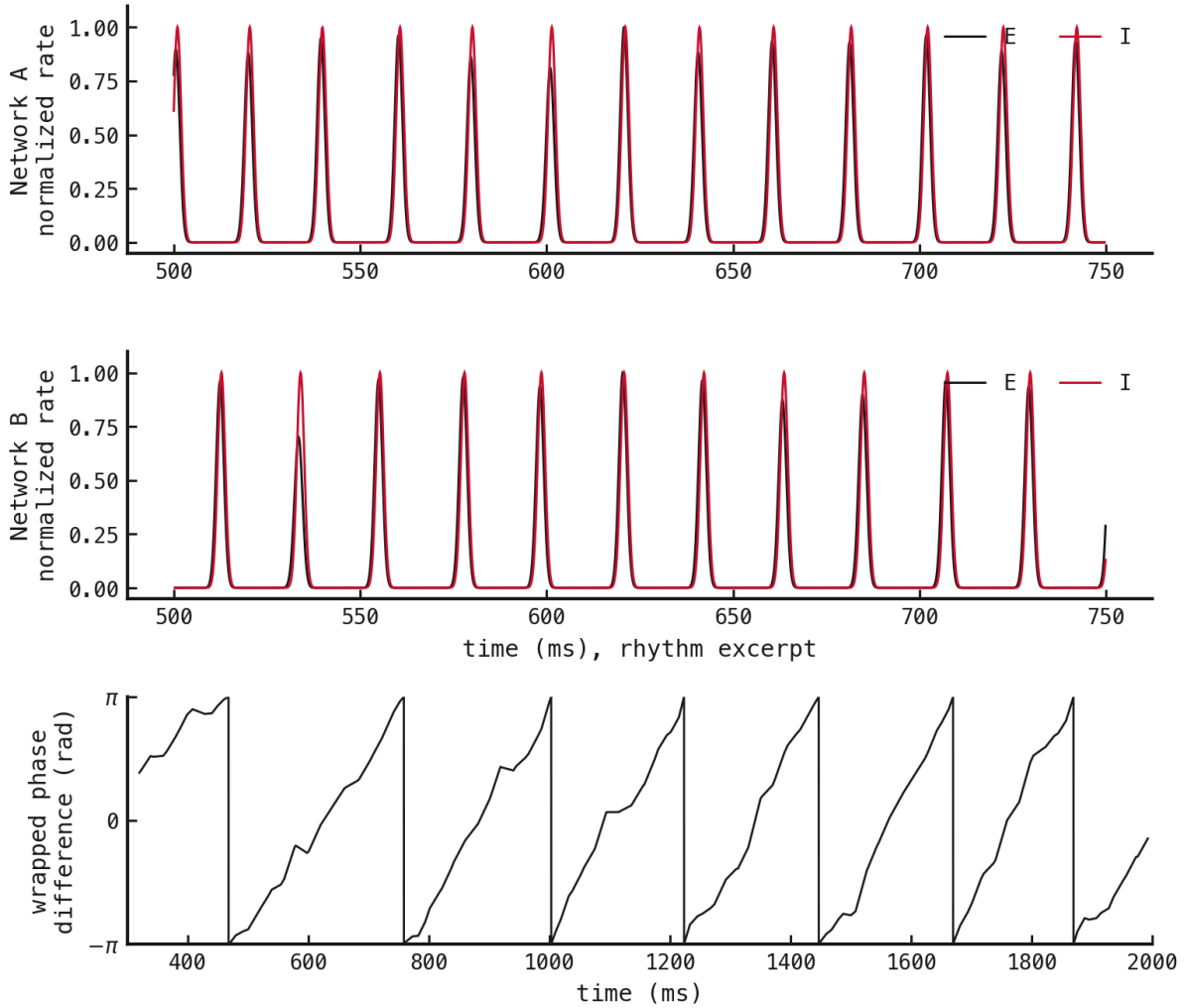


Figure 2: Uncoupled activity. The upper panels show normalized excitatory and inhibitory population rates over the same 250 ms interval. The lower panel shows the wrapped phase difference across the full post-burn recording. Vertical jumps are phase wrapping.

Network A oscillates at 50.7 Hz with inter-volley-interval CV 0.037. Network B oscillates at 46.7 Hz with CV 0.031. Every inhibitory neuron fires exactly once in every measured cycle. Both networks therefore produce clean, regular gamma rhythms without baseline inhibitory

doublets. Their relative phase wraps 7 times, establishing sustained phase drift before coupling.

Result 3: Phase-response curve

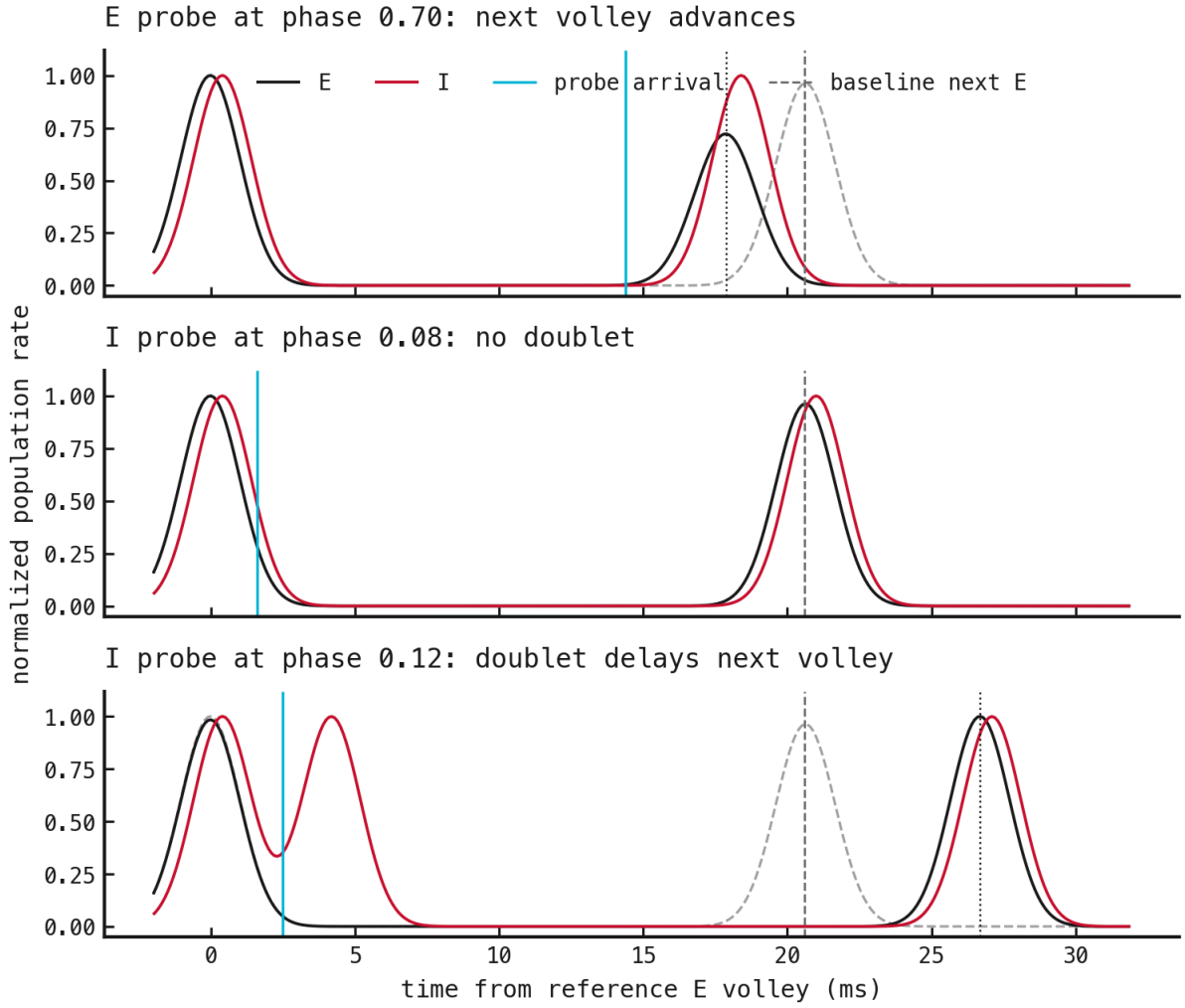


Figure 3: Representative probe responses. Time zero is the reference excitatory volley. Black and red show perturbed E and I rates. Cyan marks probe arrival. The grey dashed trace and line show the baseline E response and baseline next-volley time; a black dotted line marks a shifted next volley.

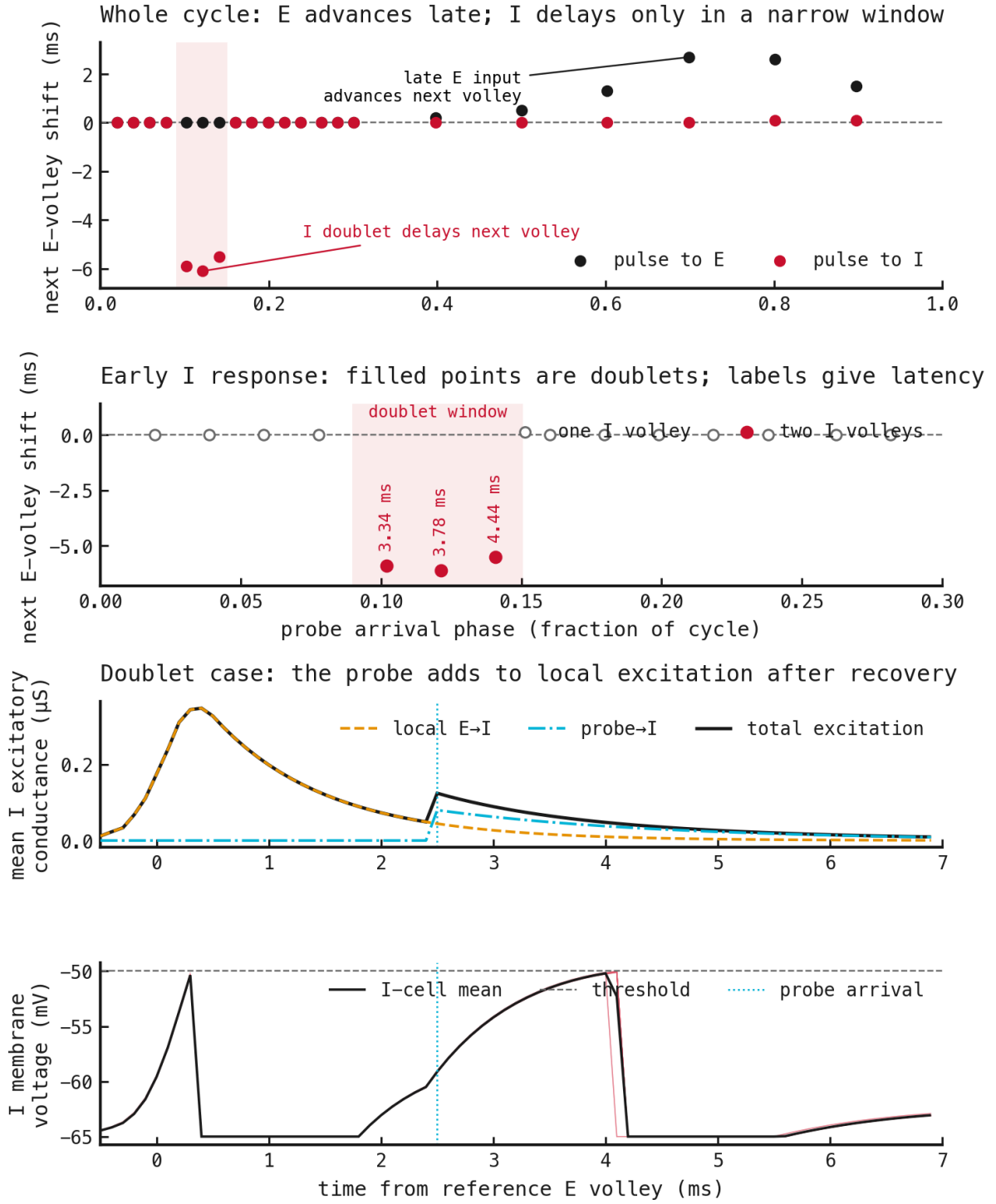


Figure 4: Response to coupling-matched probe volleys. The upper panels show the whole-cycle response and enlarged early-I window. Open points produce one inhibitory volley; filled red points produce doublets. The lower panels show local, probe, and total excitation onto I neurons and all I-cell voltages for the phase-0.12 doublet. Cyan marks probe arrival; the dashed voltage line marks threshold.

E-targeted probes advance the rhythm late in the cycle. I-targeted probes delay it only from phase 0.1 to 0.14, where they trigger a second inhibitory volley.

In the doublet case, local E-to-I conductance remains when the probe arrives. The probe raises total excitation as the recovered I-cell voltages approach threshold, producing the second volley. Earlier probes meet refractory neurons; later probes lack this remaining local excitation and are too weak alone.

Result 4: Pathway comparison

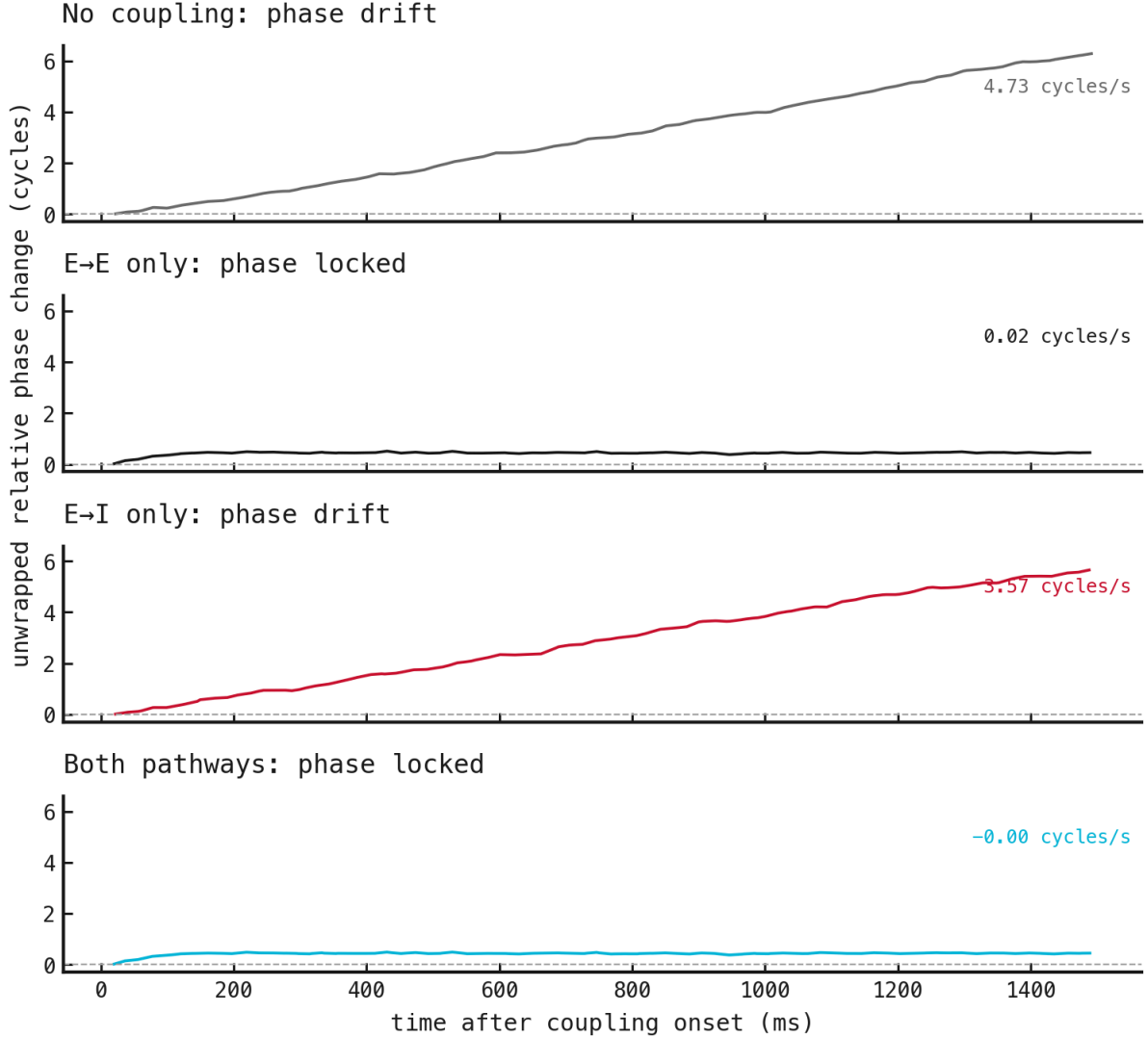


Figure 5: Pathway comparison from one shared runtime state. A continuing slope indicates phase drift; a plateau indicates phase locking. The annotation in each panel gives drift over the final 500 ms.

Without coupling, relative phase drifts at 4.73 cycles/s. E-to-I coupling alone still drifts at 3.57 cycles/s. E-to-E coupling alone reduces final drift to 0.02 cycles/s and locks the rhythms; both pathways also lock, at 0 cycles/s. Direct excitation is therefore sufficient at this operating point, whereas the narrow inhibitory-doublet pathway is not.

Result 5: Event-aligned mechanism

Axes. Time from a source excitatory volley against normalized activity or conductance.

Traces. Arriving excitation, target inhibitory activity, target inhibition, and target excitatory activity.

Why. Establish the order of events that produces each phase correction.

Expectation. The pathway responsible for locking changes before the target network's next corrected volley.